MIL-B-60135(MU)
AMENDMENT 7
30 March 1970
SUPERSEDING
AMENDMENT 6
27 January 1969

MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513599

This amendment forms a part of Military Specification MIL-B-60135(MU) dated 22 January 1965.

Paragraph 3.6.1.1: Delete paragraph and substitute:

"3.6.1.1 Operating temperatures.- There shall be no visible flicker of the image tubes while thermally stabilized at temperatures of -40°F and +125°F, when energized from an 18 to 30 vdc power source. The assembly shall be given a minimum of 5 minutes warm-up time before checking for flicker."

Paragraph 3.9: Delete paragraph and substitute:

"3.9 Image steadiness.- There shall be no image movement or visible flicker subsequent to a minimum of 5 minutes stabilization or tube warm-up time, when the assembly is energized from an 18 to 30 vdc power source or a 1.5 vdc dry battery."

Paragraph 3.11: Delete paragraph and substitute:

"3.11 Image tilt.- The image of a target line originally perpendicular to a plane, defined by the 2-inch-long shoulder (generated by the machining of the pads) adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc."

Paragraph 3.13: Delete "7.0 minutes" and substitute "4.0 minutes".

Add the following new paragraphs:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) body assemblies plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

FSC 6650

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4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to subitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.2, Table I: Delete Characteristic 103 in its entirety and re-number Characteristics 104 to 108 as 103 to 107 inclusive.

Paragraph 4.3.2: Delete "Three body assemblies" and substitute "Ten body assemblies".

Paragraph 4.3.2: Following Characteristic 306, add new requirement and test:

"307. Operating temperatures 3.6.1.1 4.5.2.1.1"

Paragraph 4.4.2.3.2: Delete paragraph and substitute:

"4.4.2.3.2 Sealing.- Special testing equipment to test the sealing of the body assembly shall conform to the requirements of Drawing F8565556, Pressure Tester-Two Station."

Paragraph 4.4.2.4: Delete and substitute:

"4.4.2.4 Devices to test optical performance requirements.- The device used to determine the acceptability of the optical requirements shall provide for the positioning of the four mounting pads in a plane perpendicular to the incident line of sight. Centrally position the body assembly in the device by locating against the 2-inch-long shoulder adjacent to the valve assembly. This device shall also provide 2 target images, each consisting of a horizontal and vertical line and appropriate resolving target pattern for resolution testing (see MIL-O-13830). The intersecting point of these lines shall be placed on the optical axis of a projector collimator. The optical axis of each collimator projector shall lie in the same plane when both vertical lines are parallel to a plumb line. Illumination of these targets (reticles) shall be by means of a Tungsten radiation source and an infrared filter. This filter shall not pass visible light rays below 0.65 microns. This testing device shall also incorporate features for holding 2 unity-power viewing telescopes arranged in a pair to be utilized as viewing collimators. The reticles of the viewing collimators shall consist

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of a horizontal and vertical centerline, the intersecting point of which shall be on the optical axis. The reticles shall also be provided with a grid pattern located on the optical axis. Each horizontal and vertical line shall represent an angle of 5 minutes. The optical axis of the viewing collimators shall be positioned perpendicular to the optical axis of the projector collimators when the vertical reticle lines are parallel to a plumb line. This device shall also provide a power supply of 24 Vdc to illuminate the body assembly under test and provide a power source to illuminate the reticles of the 2 collimators."

* Following paragraph 4.5.2.1, add new paragraph:

"4.5.2.1.1 Operating temperatures.- The body assembly shall be exposed to the temperature conditions specified in 3.6.1.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be placed in the cold chamber and the temperature of the chamber reduced gradually to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at -40°F, the body assembly shall be checked for flicker. The test shall be accomplished using the power supply capable of supplying the specified source. Apply a nominal 24 Vdc to the 24 volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. View through the eyepiece assembly of the periscope body and observe that there is no visible flicker of the image tube after being energized for the time period specified in 3.6.1.1. Upon completion of this test, the body assembly temperature shall be gradually raised to room ambient temperature. The body assembly shall then be placed into an oven chamber and the temperature shall be raised gradually to +125°F. The body assembly shall remain at this temperature for a period of 5 hours and at the conclusion of this 5 hour period, and while at +125°F, the body assembly shall again be subjected to the test depicted above to determine compliance with 3.6.1.1. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (60°F to 90°F) at which time the body assembly shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the body assembly to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.5.2.1 and 4.5.2.1.1."

Paragraph 4.5.2.3: Delete paragraph and substitute:

"4.5.2.3 Sealing.- The internal pressurizing of the body assembly to test sealing, shall be performed with special testing equipment conforming to 4.5.2.3.2. Pressurize the body assembly with dry nitrogen gas, to the value specified in 3.8 in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the body assembly for a period of 5 minutes ± seconds. At the end of the

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5 minute period there shall be no evidence of leakage, as indicated on the pressure gauge of the testing equipment, to determine compliance with 3.8. The body assembly shall then be subjected to the test in 4.5.2.4."

Paragraph 4.5.4: Delete paragraph and substitute:

"4.5.2.4 Interior atmosphere.- A dew point tester shall be used to test the dew point of the interior atmosphere of the body assembly. Upon satisfactory completion of the sealing test, the internal pressure of the body assembly shall be reduced to 1 psig. The body assembly shall then be flushed with dry nitrogen gas. The flow of gas into the body assembly shall be controlled by a regulator that will prevent the internal pressure in the body assembly from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the body assembly has a dew point at least as low as minus 25°F, as measured with the dew point tester. The body assembly shall be sealed. The internal atmosphere of the body assembly shall conform to the requirements of 3.14."

Paragraph 4.5.3.1.1: Delete paragraph and substitute the following paragraphs in sequence:

4.5.3.1.1 Image steadiness (18-30 Vdc).- Examination of body assemblies for no visible flicker or image movement as specified in 3.9, shall be performed on a 100% basis. The tests shall be performed utilizing the projector collimator inherent to the holding device. During the conduct of tests for acceptance inspection of image steadiness, the power source depicted in 4.5.3.1.1 shall be used to energize 75% of the body assemblies submitted for inspection and the power source depicted in 4.5.3.1.2 shall be used to energize the remaining 25%. The test shall be conducted using the power supply capable of supplying the specified source. Apply a nominal 24 Vdc to the 24 volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. View through the eyepiece assembly of the periscope body and observe that there is no movement of the target image or visible flicker of the image tube after being energized for the time period specified in 3.9.

4.5.3.1.2 Image steadiness (1.5 Vdc).- The test shall be performed concurrent with 4.5.3.1.1. Insert a 1.5 Vdc dry cell (C type) battery into the body assembly to energize the infrared tube. Place the double-pole double-throw switch to 1.5 volt (ON) position. Allow the time specified in 3.9 for stabilization of the target image. There shall be no visible flicker or movement of the target image to comply with 3.9.

Paragraph 6.2: Add: "(e) Requirements for first article testing."

*The outside margins have been marked to indicate where additions to the previous amendments have beem made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous issue.

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